

RIYADH KING SALMAN AB, Saudi Arabia

WMO: 404380

Lat: 24.722N

Lon: 46.725E

Elev: 635

StdP: 93.92

Time Zone: 3.00 (E03)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	8.0	-12.6	1.4	17.4	-10.1	1.7	20.8	9.5	16.9	8.8	17.5	2.0	320	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.1	44.9	19.5	44.0	19.1	43.1	18.7	21.4	37.7	20.6	37.9	19.9	38.1	4.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.9	23.0	16.8	12.9	22.1	15.2	11.6	22.2	65.2	38.3	62.2	37.3	59.8	38.0	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.2	8.2	7.2	3.4	46.3	2.1	0.9	1.9	46.9	0.7	47.4	-0.5	47.9	-2.0	48.5	
			DB												
			WB	-0.4	23.7	1.8	1.5	-1.7	24.8	-2.8	25.7	-3.8	26.5	-5.2	27.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	15.0	17.9	21.7	27.3	33.0	35.7	36.7	36.8	33.5	28.4	21.5	16.5
	DBStd	8.33	3.17	3.60	3.53	3.41	2.51	1.54	1.54	1.51	1.91	2.27	3.23	3.34
	HDD10.0	4	3	1	0	0	0	0	0	0	0	0	0	1
	HDD18.3	254	112	48	9	1	0	0	0	0	0	0	10	74
	CDD10.0	6222	157	221	362	520	712	772	827	829	704	569	344	203
	CDD18.3	3429	8	34	113	271	454	522	568	571	454	311	104	19
	CDH23.3	55228	72	316	1133	3478	7454	9289	10271	10318	7628	4203	921	145
	CDH26.7	38071	9	84	462	1900	5093	6892	7791	7838	5292	2374	317	18
Wind	WSAvg	3.2	3.1	3.6	3.6	3.8	3.5	3.6	3.5	3.2	2.7	2.4	2.8	2.9
Precipitation	PrecAvg	124	14	7	31	37	8	0	0	0	0	1	7	9
	PrecMax	318	111	38	123	132	69	0	11	6	0	27	93	70
	PrecMin	15	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	89	22	10	36	40	14	0	2	1	0	5	17	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	31.8	35.6	40.1	43.8	45.2	46.1	45.8	43.3	39.0	33.6	28.8
		MCWB	14.3	16.2	16.0	17.7	18.4	19.0	19.9	20.2	19.0	17.3	16.7	15.4
	2%	DB	25.2	29.2	33.2	38.1	42.2	44.1	45.0	44.6	42.2	37.4	31.9	26.9
		MCWB	13.3	14.6	15.6	17.4	18.3	18.7	19.4	19.7	18.7	16.7	15.7	14.7
	5%	DB	23.2	27.1	31.2	36.8	41.0	43.1	44.0	44.0	41.2	36.2	30.2	24.9
		MCWB	12.2	13.9	14.9	16.9	18.2	18.4	18.9	19.4	18.3	16.4	15.3	13.7
	10%	DB	21.2	25.0	29.2	34.8	39.9	42.1	43.0	43.1	40.1	35.1	28.5	23.0
		MCWB	11.8	12.8	14.1	16.6	18.0	18.0	18.6	19.1	17.8	16.3	14.9	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	17.7	19.7	21.0	21.6	20.6	22.3	23.5	21.1	20.2	19.9	18.6
		MCDB	21.6	24.5	26.3	28.5	32.4	41.8	39.5	37.2	35.6	27.6	25.1	23.3
	2%	WB	15.7	16.5	17.8	19.8	20.3	19.6	20.8	21.6	19.8	18.8	18.5	17.1
		MCDB	20.5	24.1	26.3	29.3	35.0	42.0	41.6	40.4	39.1	30.3	23.9	21.2
	5%	WB	14.5	15.3	16.7	18.9	19.5	19.0	19.8	20.7	19.1	17.9	17.5	15.9
		MCDB	20.1	23.9	25.6	29.4	35.9	41.7	42.0	40.6	39.1	32.1	24.1	21.7
	10%	WB	13.2	14.1	15.7	18.0	18.8	18.4	19.1	19.9	18.5	17.2	16.5	14.5
		MCDB	19.0	23.2	25.1	30.0	36.2	41.1	41.9	41.0	38.5	32.5	24.6	20.8
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.4	11.9	12.1	12.5	13.1	12.9	13.1	13.2	13.0	11.1	10.7
		MCDBR	13.4	13.7	14.6	14.1	13.6	13.9	13.4	13.3	13.7	13.9	13.3	13.3
		MCWBR	6.4	6.0	6.1	5.1	4.6	4.7	4.6	4.6	4.8	5.2	5.5	6.1
	5% WB	MCDBR	10.3	11.7	11.5	11.9	11.9	13.0	12.5	12.3	12.7	11.9	9.8	9.6
		MCWBR	6.0	5.8	5.9	4.8	4.2	4.6	4.7	4.6	4.6	5.1	4.4	5.1
Clear-Sky Solar Irradiance	taub		0.427	0.471	0.541	0.622	0.644	0.603	0.583	0.574	0.517	0.450	0.461	0.423
	taud		1.980	1.854	1.697	1.572	1.531	1.586	1.655	1.675	1.771	1.923	1.904	1.990
	Ebn at Noon		809	802	767	717	700	724	738	744	776	810	765	792
	Edh at Noon		160	195	240	277	288	271	253	247	219	179	171	153
All-Sky Solar Radiation	RadAvg		4.31	5.16	5.93	6.45	7.19	7.83	7.48	7.12	6.55	5.71	4.42	4.08
	RadStd		0.30	0.29	0.33	0.32	0.28	0.24	0.17	0.16	0.15	0.21	0.31	0.27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	+0.87	N/A	N/A	N/A	N/A	+250	+260
Regional (1 neighbor)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+103

Nomenclature: See separate page